HAMK
Häme University
of Applied Sciences

Microsoft based performance goal management system for strategy implementation

Bachelor's Thesis

Degree Programme in Business Information Technology

Spring 2026

Sawsan Abdulbari

HAMK
Häme University of Applied Sciences

Tiivistelmä

Koulutus	Tietojenkäsittelyn koulutus	
Tekijä	Sawsan Abdulbari	Vuosi 2026
Työn nimi	Microsoft based performance goal management system for strategy implementation	
Ohjaaja	Kevin Cheng	

Tämän opinnäytetyön tarkoituksena oli luoda ja toteuttaa Microsoft-pohjainen tavoitteidenhallintajärjestelmä Kiipulasäätiö sr:lle, joka on etelä-Suomessa toimiva koulutus- ja kuntoutustoimintaan keskittyvä suomalainen säätiö. Opinnäytetyössä käsiteltiin Excel-pohjaisen tavoitteiden seurannan aiheuttamia ongelmia, jotka johtivat läpinäkyvyyden puutteeseen, tietojen epäjohdonmukaisuuteen sekä yrityksen tavoitteiden ja päivittäisten tehtävien välisen yhteyden heikkenemiseen. Päätavoitteena oli rakentaa kestävä digitaalinen ratkaisu, jossa pääalustana toimi SharePoint Online.

Opinnäytetyön tavoitteena oli selvittää, kuinka SharePoint-pohjainen järjestelmä voi korvata Excel-pohjaiset tavoitteidenhallintakäytännöt strukturoidulla ja kestävällä tavalla, kuinka suunniteltu järjestelmä voi tukea data-analytiikkaa ja tieteelliseen päätöksentekoon suuntautuvaa johtamista organisaatioympäristössä, sekä kuinka integroitu Microsoft-pohjainen ratkaisu parantaa läpinäkyvyyttä, tietojen laatua ja analyyttistä käytettävyyttä verrattuna sirpaloituneisiin Excel-ratkaisuihin.

Työ oli toiminnallinen ja painottui vahvasti kehitystyöhön. Se yhdisti teoreettisia käsitteitä, kuten tavoitteenasettamisteoriaa, Balanced Scorecard -viitekehystä, tietojen laatuun liittyviä periaatteita sekä digitaalisen työpaikan ideoita, iteratiiviseen suunnitteluun ja järjestelmän toteutukseen.

Intranetille toteutettu pilvipohjainen ratkaisu voi yhdistää strategian ja operatiivisen toiminnan varmistamalla yhtenäisten tietomallien käytön SharePoint-listoilla ja hakuarvoilla. Ratkaisu mahdollistaa tavoitteiden allokoinnin organisaation eri tasoille Power Automate -automaatioiden avulla. Se tarjoaa reaaliaikaisen vuorovaikutuksen Power Apps -sovelluksella ja tuo tulokset näkyviin Power BI:ssä ehdollisen muotoilun, trendianalyysin ja porautumismahdollisuuksien avulla. Toteutettu ratkaisu on parantanut tietojen tarkkuutta ja vähentänyt manuaalista työtä samalla tukien johdon päätöksentekoa. Testauksen aikana toimeksiantajilta saatu palauteen mukaan ratkaisu on helposti käytettävä ja opittava. Henkilötieto hakemiston rajoitusten vuoksi tiimien valinta tapahtui manuaalisesti personoinnin sijaan.

Avainsanat	Tavoitteidenhallinta, SharePoint, Power BI, strategian toimeenpano, digitaalinen työpaikka.
Sivut	28 sivua ja liitteitä 3 sivua

HAMK
Häme University of Applied Sciences

Abstract

DP	Degree Programme in Business Information Technology	
Author	Sawsan Abdulbari	Year 2026
Subject	Microsoft based performance goal management system for strategy implementation	
Supervisors	Kevin Cheng	

The purpose of this bachelor's thesis was to create and implement a Microsoft-based goal management system for Kiipulasäätiö sr., a Finnish foundation focused on education and rehabilitation in southern Finland. The thesis addressed the problem with Excel-based goal tracking, which led to issues with transparency, consistency of the data, and the connection between company goals and daily tasks. The main objective was to build a structured and sustainable digital solution using SharePoint Online as the primary platform.

The thesis aimed to find answers to how a SharePoint-based system can replace Excel-based goal management in a way that is structured and sustainable, how the designed system can support data analytics and decision-making in an organizational context, as well as how an integrated Microsoft-based solution could be more transparent, have better data quality, and analytical usability compared to Excel.

The work was functional and focused heavily on the development of a new goal management system using Microsoft tools. The development was based on many theoretical ideas like goal-setting theory, the Balanced Scorecard framework, rules for data quality, and digital workplace ideas.

The research showed that a cloud solution for the intranet can connect the strategy with everyday operations by the use of consistent data models with SharePoint lists and lookup values. The solution can share goals with various levels of the organization using Power Automate. The solution can provide real-time role-specific interaction with Power Apps and can bring about the results using Power BI with conditional formatting, trend analysis, and drill-down features. The solution implemented has ensured better data accuracy and reduced manual work while supporting management decisions with evidence. Feedback from the commissioners during the test has indicated that the solution is easy to use with minimal training required. However, limitations in the directory data meant manual team selection instead of personalization.

Keywords	Goal management, SharePoint, Power BI, strategy implementation, digital workplace.
Pages	28 pages and appendices 3 pages

Glossary

Balanced Scorecard (BSC)	A strategic management framework that translates organizational strategy into objectives from financial, customer, internal process, and learning and growth perspectives.
DAX (Data Analysis Expressions)	A formula language in Power BI used to create custom calculations and measures for dynamic data analysis.
Digital Workplace	A modern intranet-based environment that integrates information access, collaboration, automation, and analytics to support daily work.
Goal Alignment	The process of linking individual, team, departmental, and organizational goals to achieve coherence with the overall strategic direction.
Goal-Setting Theory	A framework that defines specifics like challenging goals, combined with feedback and commitment, grows individual and organizational performance.
Information Architecture	The structural design of shared information environments to support usability, findability, and navigation in digital systems.
Intranet	An internal organizational network with options for communication, collaboration, document management, and data access.
Levers of Control	A management framework using belief systems, boundary systems, diagnostic controls, and interactive controls to guide strategy implementation and risk management.

Metadata	Descriptive data assigned to items (e.g., in SharePoint lists) to enable classification, search, filtering, and multi-dimensional analysis.
Microsoft 365	A cloud-based productivity suite (including SharePoint, Power BI, Power Apps, and Power Automate) that supports collaboration, automation, and analytics.
Power Apps	A low-code platform for building custom canvas or model-driven applications that integrate with SharePoint data and provide interactive user interfaces.
Power Automate	A workflow automation service that creates flows to automate tasks, such as goal propagation across SharePoint lists.
Power BI	Microsoft's business analytics tool for transforming data into interactive visualizations, reports, and dashboards.
SharePoint List	A structured data container in SharePoint with defined columns, validation rules, and views for managing and displaying information like goals or teams.
SharePoint Online	The cloud-based version of SharePoint within Microsoft 365, providing sites, lists, libraries, pages, and integration with other Power Platform tools.
User-Centered Design	An approach to system development that prioritizes usability, feedback, error prevention, and consistency to reduce cognitive load and improve adoption.

Table of Contents

1 Introduction 1
2 Goal management and strategy implementation 2
2.1 Strategic goal management in organizations 2
2.1.1 Foundations of goal-setting theory 3
2.1.2 The gap between strategy and operations 3
2.1.3 Organizational goal alignment 4
2.2 Balanced scorecard as a strategic management framework 5
2.2.1 The Balanced Scorecard as a tool for goal alignment 5
2.2.2 The four BSC perspectives 5
2.3 Goal data management and organizational information systems 6
2.3.1 Limitations of spreadsheet-based goal management 6
2.3.2 Information architecture and user-centered design 6
2.3.3 Information architecture and user-centered design 7
2.3.4 Collaboration and digital work environments 7
3 Intranet systems and SharePoint as a platform 7
3.1 Digital workplaces and organization intranets 8
3.2 Microsoft SharePoint architecture and capabilities 9
3.3 SharePoint lists, metadata and structured data management 10
3.4 Power BI components and data modelling 11
3.5 Effective visualisation and dashboard design 13
4 Methods 13
4.1 Research approach 14
4.2 Development methods 14
4.3 Data collection 14
4.4 Process of designing the system 15
4.5 Evaluation methods 15
5 Practical implementation of the goal management system 16
5.1 System architecture and platform selection 17
5.2 Data structure and SharePoint lists configuration 18
5.3 Automated goal propagation logic 19
5.4 User interface design and navigation structure 19
5.5 Visual representation of goals 21
5.6 Visualization design for general users 22

5.7 Visualization design for managerial oversight 23
5.8 Integration of analytics into the intranet environment 24
5.9 Access control and security design 24
5.10 Iterative development and validation 24
6 Results 25
7 Summary 26
References 28

Figures

Figure 1. Kiipula goal management system diagram 17
Figure 2. View of adding an entry to the organizational goals list in SharePoint 18
Figure 3. Power Automate flow for automated goal entry propagation 19
Figure 4. Sector page buttons on the home page of the goal management system 20
Figure 5. The sector and team selection dropdowns 21
Figure 6. Team specific goal display provided when team was selected 22
Figure 7. General workers Power BI report 22

Program codes

Program code 1. DAX expressions calculating the total amount of goals completed 12

Appendices

Appendix 1. Data management plan

1 Introduction

Organizations aim to convert strategic goals into specific and measurable operational activities to be successful in the long run. Practically, this is not an easy task because of disjointed tools which are not easily integrated with daily work or help to conduct systematized analysis. The lost connection between strategic planning and operational implementation normally lead to inefficiencies, lack of transparency, and the inability to make decisions based on data.

As a student specializing in data science and machine learning at HAMK University, I am interested in the subject because I have personally experienced the constraint of poor data infrastructure in the context of restricting analytical processes, despite an organization having competent professionals and well-defined strategic objectives. From data science perspective, solutions based on Excel are readily available and familiar, but they limit the creation of a functioning data ecosystem by creating inconsistent data models, data collection practices that are fragmented, and have limited analytical scale.

Kiipulasäätiö sr, Finnish organization in the field of education and rehabilitation, exemplifies this challenge. The organization needs systems which can coordinate operations of multiple departments, programs, and stakeholders. Despite clear strategic targets, the use of Excel sheets to manage team level goals has proven to be problematic. Such spreadsheets are hard to sustain and scale, have uneven data structures and cannot be used analytically throughout the organization. The current situation has given reason to find a more formalized, comprehensive and analytical solution.

This thesis aims to create and deploy Microsoft intranet goal management system for Kiipulasäätiö sr. Development of machine learning models is not in the scope of the thesis.

This thesis addresses three research questions

- How can SharePoint-based system replace Excel-based goal management practices in structured and sustainable way?
- How can the designed system support data analytics and science-oriented decision-making in an organizational context?
- How does an integrated Microsoft-based solutions improve transparency, data quality, and analytical usability compared to fragmented Excel solutions?

2 Goal management and strategy implementation

Effective strategy execution and organizational performance are based on strategic goal management (Simons, 1994, p. 4). The chapter provides a review of some of the important theoretical views associated with goal management, strategy implementation, and organizational information systems (Mintzberg, 1994, p. 112). The discussion sets the conceptual framework of the design and implementation of the goal management system designed in the current thesis and puts the work in context with the other strategic management, performance measurement, and data-driven organizational decisions (Popovič et al., 2012, p. 729; Simons, 1994, p. 4).

2.1 Strategic goal management in organizations

Strategic goal management is the methodical way of transforming organizational vision and strategy into tangible and quantifiable goals that can be used to direct the day-to-day operations and decision making (Simons, 1994, pp. 7–8). Though most organizations can develop elaborate strategies, the effective implementation of these strategies has continued to be a challenge (Mintzberg, 1994, p. 112). According to previous studies, failure of strategy implementation is by far much more frequent than strategy formulation (Verma & Sharma, 2016, p. 2).

The main challenge is the ability to close the gap between the high-level strategic intent and operational reality (Verma & Sharma, 2016, p. 1). Strategic goals are usually created in abstract language that can hardly be translated into something that could be implemented by teams and individuals (Mintzberg, 1994, p. 107). The everyday choices can become disconnected with strategic priorities, lowering the organizational effectiveness (Simons, 1994, p. 8). Good goal management systems strive to minimise this gap by establishing strong connections between strategy, goals, measures of performance and feedback systems (Simons, 1994, p. 4).

Empirical research conducted by Kaplan and Norton (2001) has shown that organizational strategy is often not well known to employees and the small percentage of the workforce can describe strategic priorities and how their roles are involved in their achievement. Meanwhile, management systems tend to be short-term oriented in terms of financial measures instead of strategic goals and resource allocation process is not linked with strategic priorities. These do not rule out the importance of effective goal management

systems that incorporate both strategy and operational implementation (Kaplan & Norton, 2001, p. 3).

2.1.1 Foundations of goal-setting theory

The goal setting theory offers a psychological and behavioural basis on studying the role of goals in performance (Locke & Latham, 2002). The theory was developed based on the many experiments conducted by Locke & Latham (2002, p. 706) and shows that with specific and challenging goals, there is an increase in the level of performance as compared to vague ones. Goals control behaviour by taking focus, motivating effort, making it more persistent and encouraging the generation of effective strategy of doing tasks (Locke & Latham, 2002, pp. 705–706).

Goal setting is effective depending on a number of moderating factors (Locke & Latham, 2002). Goal commitment is what defines the motivation of people to work on goals and self-efficacy is what defines confidence in attaining the goals. Feedback is also very important as it helps individuals and groups to evaluate progress and make necessary changes. Even clear goals fail to inspire without fast and proper feedback (Locke & Latham, 2002, p. 707).

Goal-setting theory in the context of organizations is not only restricted to the individual performance but also to the team and organizational performance (Kaplan & Norton, 2001). Goal management demands coordination between levels, and this implies that individual and team goals must be tied to departmental and organizational goals (Kaplan & Norton, 2001, p. 195). Such alignment produces a coherent system whereby strategic priorities are shown throughout the process of operation (Locke & Latham, 2002, p. 709).

2.1.2 The gap between strategy and operations

Strategy-operations gap is a term that is used to explain the gap between the strategic plans that top management come up with and the actual reality of operations that are faced by the employees (Kaplan & Norton, 2001). According to Kaplan & Norton (2001, pp. 5–6), there are four main obstacles that cause this gap.

The vision barrier comes when the strategy is not communicated effectively, and the employees have less understanding of it (Kaplan & Norton, 2001). People barrier is

expected when there is no alignment between incentives, performance review and skill improvement and strategic priorities. The resource barrier indicates inability to connect budgets with resource allocation to strategy so that the short-term operational needs can take priority over long-term strategic projects. Management barrier arises when the executives do not give enough time to strategic discussion and instead concentrate on the operations and financial reporting (Kaplan & Norton, 2001, pp. 10–11).

These obstacles combine to ensure strategy remains a disjointed and discrete component of the management of the organization (Kaplan & Norton, 2001, pp. 11–12). To mitigate the strategy operations gap, it is necessary to have systems that entrench strategy into the daily operations, performance measurement and decision-making frameworks (Kaplan & Norton, 2001, pp. 12–13).

2.1.3 Organizational goal alignment

Organizational goal alignment is the extent to which organizational goals on various levels and functions aid a shared strategic direction (Kaplan & Norton, 2001, p. 192). There are three dimensions of alignment, namely vertical, horizontal, and temporal (Kaplan & Norton, 2001, pp. 161–163).

Vertical alignment makes goals flow in a logical way within the organizational strategy, departments, teams and individuals (Kaplan & Norton, 2001). The levels translate the higher-level goals into more specific and actionable goals without losing a distinct linkage to strategic priorities. Horizontal alignment deals with coordination of functional units and eliminates conflicts and duplication of efforts. Temporal alignment synchronizes the short-term operational goals with the long-term strategic goals in such a way that the short performance demands do not compromise on the long-term performances (Kaplan & Norton, 2001, pp. 161–163).

Goal management systems should be able to underpin all three aspects of alignment at the same time (Kaplan & Norton, 2001, p. 165). Data wise it involves the use of consistent goal structures, common definitions and integrated data models that enable goals to be seen and analyzed across organizational boundaries and time (Kaplan & Norton, 2001, p. 170).

2.2 Balanced scorecard as a strategic management framework

The balanced scorecard (BSC) offers a popular model of strategy expression into operational goals and performance indicators (Kaplan & Norton, 2001). The BSC is used to overcome the shortcomings of the traditional financial performance measurement systems, since it includes non-financial measures of customer, internal processes and organizational learning (Kaplan & Norton, 2001, pp. 2–3).

2.2.1 The Balanced Scorecard as a tool for goal alignment

The balanced scorecard has the short term and the long-term goals, financial and non-financial measures, leading and lagging indicators (Kaplan & Norton, 2001). Instead of defining strategy as a plan, the BSC conceptualizes strategy as a set of assumptions about the creation of value. Performance measurement and feedback can be used to test and refine it after (Kaplan & Norton, 2001, pp. 6–8).

Kaplan & Norton (2001, p. 3) also expanded the BSC into a strategic management system focusing on the translation of strategy into operational language, alignment of the organization, strategic learning and leadership at the executive level. Strategy maps are of key importance as they visually depict cause-effect relationships among strategic objectives (Kaplan & Norton, 2001, p. 72).

2.2.2 The four BSC perspectives

The BSC categorizes strategic objectives into four perspectives which include financial, customer, internal process as well as learning and growth (Kaplan & Norton, 2001). All perspectives cover a different aspect of organizational performance and are linked to each other.

The financial perspective assesses the contribution of strategy to intended results of the economy. Customer perspective deals with the creation of value to target stakeholders. Internal process perspective determines key processes that facilitate strategy implementation. The learning and growth view focuses on the ability, mechanisms and culture to maintain long-term performance.

In the case of non-profit and government agencies, including social services organizations, the BSC model needs to be modified. Financial performance is frequently taken over by mission achievement as the main goal, and financial considerations are taken as a limitation to goals (Kaplan & Norton, 2001, pp. 75–77, 133–135).

2.3 Goal data management and organizational information systems

It is not only important that goal management is based on conceptual frameworks but also on information systems that are able to sustain data collection, analysis, and communication (Popovič et al., 2012, p. 737). The development of the paper-based systems to spreadsheets and eventually to integrated digital platforms indicates the growing needs of data quality, accessibility and analytics (Popovič et al., 2012, p. 731).

2.3.1 Limitations of spreadsheet-based goal management

Goal tracking systems based on spreadsheets like the Excel have been in use long before because they are flexible and inexpensive to start (Panko, 2005, p. 2). Nevertheless, studies have found considerable weaknesses in the utilization of spreadsheets, such as the fact that they have problems with data integrity, version control, lack of collaboration facilities, and challenge in scaling (Panko, 2005, pp. 3–4).

Separated and inconsistent files, ad hoc working style, and manual processes do not allow systematic analysis and also it makes data less trustworthy (Popovič et al., 2012, p. 731). This kind of environment is not fit for advanced analytics and machine learning as they do not have the standardized data pipelines which would secure them (Panko, 2005, p. 5; Popovič et al., 2012, p. 731).

2.3.2 Information architecture and user-centered design

The quality of data is a very important factor that defines the utility of goal management systems (Wang & Strong, 1996, p. 5). According to the data quality framework by Wang & Strong (1996, pp. 21–22), these are intrinsic, contextual, representational, and accessibility dimensions. If the quality of data is not good, no insight in analysis or decisions can be made, no matter the tools of analysis (Wang & Strong, 1996, pp. 21–22).

Integrated solutions like SharePoint overcome most issues of data quality by imposing structured data models, real time access, version control, and role-based access control (Microsoft 365 Documentation, n.d.). These aspects provide a base on credible analytics and fit data science applications (Wang & Strong, 1996, pp. 9–10).

2.3.3 Information architecture and user-centered design

The principles of information architecture and user-centered design are very important to system usability and adoption, Few (2006) found. Goal data can be easily interacted with by providing clear organization, regular labeling, easy navigation, and excellent search capabilities. The systems should support the various user requirements such as executives who need top-level dashboards and team members reporting on operational progress.

According to Few (2006), user-centered design focuses on system status visibility, feedback, preventing errors, and consistency. These rules decrease the cognitive load and promote regular use which makes data quality better and the reliability of the analysis (Few, 2006, pp. 12–13, 45, 106–107, 139).

2.3.4 Collaboration and digital work environments

The recent goal management software tends toward collaborative functions including shared workspaces, notifications, comments, and communication tools integration (Williams & Schubert, 2018). These features acknowledge the fact that goal achievement is a team effort that necessitates coordination, transparency, and common understanding.

In systems such as SharePoint, integration with applications, such as Microsoft Teams, allows smooth cooperation between the units of an organization (Microsoft 365 Documentation, n.d.). In the case of organizations whose operations are disrupted, like Kiipulasäätiö sr, the capabilities provide coherent goal management and local autonomy (Williams & Schubert, 2018, pp. 478–479).

3 Intranet systems and SharePoint as a platform

Organizational intranets have transformed into cohesive digital workspaces that assist in structured information management, teamwork, and work based on data (Williams & Schubert, 2018, p. 479). Instead of being the repositories of information, modern intranets

are organizational information infrastructure that bridge day to day operations and communication and analytics (Baptista et al., n.d., p. 12). The chapter discusses Microsoft SharePoint as a cloud-based data structure management tool and the use of lists, metadata, and information architecture in making data analytics ready and Power BI as a built-in analytical layer facilitates data transformation, visualization, and organizational decision-making (Microsoft 365 Documentation, n.d.).

3.1 Digital workplaces and organization intranets

Intranets in organizations have changes significantly in the latter years (Baptista et al., n.d.). Early intranets were internal websites that were mostly used to disseminate unchanging information like policies, procedures, and announcements. Such systems were used to facilitate a one-way communication between the organization and the employees and were not interacting or customized. Even though they enhanced accessibility to communication as compared to paper-based communication, their strategic value was minimal.

Later intranets were developed with collaboration elements, like document sharing, discussion forums and user generated material, describes Baptista et al. (n.d.). Although these second-generation intranets grew participation, most organizations had challenges associated with ineffective information architecture, inconsistent quality of content and inability to integrate with work processes. Content became outdated and fragmented over the time which resulted in decreasing user engagement and less confidence in the intranet as a reliable information (Baptista et al., n.d., pp. 7, 12–13, 22–23).

Modern intranets that are sometimes called workplaces are a third step in evolution that is facilitated by cloud platforms and social technologies (Williams & Schubert, 2018, p. 479). The current intranets have customized user experiences, are easily compatible with productivity software, and facilitate cross-organizational and geographical collaboration (Williams & Schubert, 2018, p. 483). Williams and Schubert (2018, p. 479) argue that successful digital workplaces can be used as the main source and environment for information, work, collaboration with coworkers, and analytics based on worker roles. These systems can also always be developed in terms of mobile accessibility, real-time updates, and built-in analytics using automations (Baptista et al., n.d., p. 19; Williams & Schubert, 2018, pp. 480–481).

The strategic value of the modern intranets is not limited to information dissemination as mentioned by Williams and Schubert (2018). They facilitate organizational communication by allowing information to flow up and down, improve collaboration via shared workspaces and social capabilities, and automate as well as enable self-service capabilities. In the data viewpoint, intranets are becoming a source of operational data whereby structured information is contained in the intranets that can be analysed to aid in management and decision making (Williams & Schubert, 2018, pp. 479–480, 483).

According to Baptista et al. (n.d.), intranet implementations have continued to be challenged despite their potential. It has been found that executive sponsorship, governance, and content strategy are key to maintaining intranet value in the long run. In the absence of proper ownership and maintenance procedures, intranets are susceptible to content degradation, lack of findability, and failure to be adopted. The design and constant assessment of the users are thus vital to make sure that the intranets are aligned to organizational requirements and the changing work practice (Baptista et al., n.d., pp. 16–19, 28–30).

3.2 Microsoft SharePoint architecture and capabilities

Microsoft SharePoint is one of the most popular intranet and collaboration solutions in the world, having hundreds of millions of users in a publicly, privately, and non-profit organizations (Microsoft 365 Documentation, n.d.). In the beginning SharePoint was developed as an on-premises server product but has since become a cloud-native service in the Microsoft 365 ecosystem (Microsoft 365 Documentation, n.d.). This development indicates the overall trends in enterprise information systems that are moving towards software-as-a-service models that are more focused on scalability, integration, and continuous improvement (Microsoft 365 Documentation, n.d.).

The SharePoint online runs on the global cloud infrastructure provided by Microsoft and transfers the infrastructure management, security patching, scalability, and availability to the service provider (Microsoft 365 Documentation, n.d.). This model of architecture minimizes the technical overhead of organizations and allows organizations to roll out quickly and update features continuously (Microsoft 365 Documentation, n.d.). As opposed to the traditional software systems where periodic upgrade projects are in place, SharePoint online is constantly updated to add new features and security improvements with time (Microsoft 365 Documentation, n.d.). This type of continuous delivery model

promotes long-term sustainability but necessitates organizations to change the governance practices in order to cope with the change (Microsoft 365 Documentation, n.d.).

Architecturally, SharePoint structures the contents in a hierarchical manner comprising of tenants, site collections and sites (Microsoft 365 Documentation, n.d.). It is the organizational boundary of the Microsoft 365, which includes users, security, and global settings (Microsoft 365 Documentation, n.d.). Site collections give logical and security boundaries of content and applications within it, and pages, lists, and document libraries are enclosed by separate sites (Microsoft 365 Documentation, n.d.). The main difference between modern SharePoint and the previous version lies in the fact that team sites are used in the context of collaboration within specific groups, whereas the communication sites are used to spread information to wider audience (Microsoft 365 Documentation, n.d.).

SharePoint pages are modular canvases that are made up of web parts that show text, media, lists, documents, and embedded applications (Microsoft 365 Documentation, n.d.). Document libraries are structured document management systems that have version control, metadata, and workflow support (Microsoft 365 Documentation, n.d.). SharePoint list, the one that is especially applicable to the thesis, provide pliable structured data storage that fills the divide between straightforward spreadsheets and relational databases (Microsoft 365 Documentation, n.d.). Collectively these elements enable SharePoint to operate as an informational system as opposed to a document storage facility (Microsoft 365 Documentation, n.d.).

3.3 SharePoint lists, metadata and structured data management

SharePoint lists offer a flexible system of storing and manipulating organized organizational information (Microsoft 365 Documentation, n.d.). Every list is defined as a schema with columns of defined data types, validation rules, and user interface controls (Microsoft 365 Documentation, n.d.). Types of available column are text, numeric values, dates, choices, and Boolean values and user or other list references (Microsoft 365 Documentation, n.d.). This embedded system of type eliminates data inconsistencies and is user friendly to non-technical users (Microsoft 365 Documentation, n.d.).

Compared to spreadsheets where the structure and editing of the data can be done freely, SharePoint lists are limited in their structure to facilitate quality data and long-term flexibility

(Microsoft 365 Documentation, n.d.). Mandatory fields make the form complete, calculated columns and validation rules minimize the errors in data entry (Microsoft 365 Documentation, n.d.). List views enable one underlying data to be displayed in different ways based on user requirements so that it can be filtered, grouped or summarized without the need to duplicate data (Microsoft 365 Documentation, n.d.). This data storage and presentation separation is an essential part of analytical preparedness (Microsoft 365 Documentation, n.d.).

Metadata is very important in augmenting the worth of share point lists and documents (Microsoft 365 Documentation, n.d.). Metadata allows a flexible classification, better searchability, and multi-dimensional analysis by assigning descriptive properties to items (Microsoft 365 Documentation, n.d.). SharePoint has simple metadata columns and managed metadata based on centralized taxonomies (Microsoft 365 Documentation, n.d.). Managed metadata provides a consistency throughout the organization (Microsoft 365 Documentation, n.d.). To manage goals, metadata may be used to standardize concepts of strategic themes, organizational units, goal status, and periods of time, building a comprehensive data model across teams and departments (Microsoft 365 Documentation, n.d.).

In the context of data science, reliable analytics is based on structured lists and metadata (Popovič et al., 2012, pp. 736–737). SharePoint lists can be used in the development of stable data pipelines that can be visualized and used in machine learning in the future unlike spreadsheet-based systems, which are often characterized by fragmentation and inconsistency (Microsoft 365 Documentation, n.d.).

3.4 Power BI components and data modelling

Microsoft has developed the Power BI as its business analytics tool that converts raw data into interactive graphs (Microsoft 365 Documentation, n.d.). Power BI would facilitate the analysis of operational data obtained in lists and libraries without having to duplicate and merge data sources when combined with SharePoint (Microsoft 365 Documentation, n.d.). This interconnection facilitates a layered architecture where SharePoint and Power BI are used as operational and analysis data stores, respectively (Microsoft 365 Documentation, n.d.).

Power BI is a combination of multiple inter-connected elements (Microsoft 365 Documentation, n.d.). The first tool in the Power BI toolkit is the Power BI Desktop which is the main authoring tool used to connect data sources, transform data, define data models and generate reports (Microsoft 365 Documentation, n.d.). Published reports and datasets are stored in the Power BI Service, which allows sharing, collaboration, automatic data update, and access control (Microsoft 365 Documentation, n.d.). Embedding of reports can be done directly in SharePoint pages, which enables users to take in analytics in the same environment that they use in handling operational data (Microsoft 365 Documentation, n.d.).

One of the key modules of Power BI is Power Query, a data preparation extract-transform-load tool (Microsoft 365 Documentation, n.d.). Power query allows users to be connected to SharePoint lists, eliminate redundant system fields, normalize data types and perform data transformations, including filtering, aggregation, and reshaping (Microsoft 365 Documentation, n.d.). These transformation steps are stored and are reused automatically in the case data refresh, so that they are consistent and reproducible (Microsoft 365 Documentation, n.d.). The graphical interface can satisfy most of the requirements so advanced users can extend transformations with M language (Microsoft 365 Documentation, n.d.).

Once data is loaded, the data model in Power BI establishes a relationship between tables and allows computing of Analyses using data Analysis Expressions (DAX) (Microsoft 365 Documentation, n.d.). The measures developed using DAX make dynamic computations possible in response to filters and user interactions (Microsoft 365 Documentation, n.d.). As an example, the completion rates of goals, the number of overdue goals, time-related trends can be calculated with DAX expressions, like the Program code 1 for calculating the total amount of completed goals.

Program code 1. DAX expressions calculating the total amount of goals completed

```
Completion rate = DIVIDE(COUNTROWS(FILTER(Goals, Goals[Status] =
“Completed”)), COUNTROWS(Goals), 0)
```

These measures facilitate real-time analytical exploration and allow defining key performance indicators in a consistent manner across reports (Microsoft 365 Documentation, n.d.). Intelligence of time also enables the analysis of trends, performance during a year, and comparison between periods in case an appropriate date dimension is added (Microsoft 365 Documentation, n.d.).

3.5 Effective visualisation and dashboard design

According to Few (2006), to move from analytical findings into implementing it in practice, effective data visualization is a must. Dashboards that are developed should then be clear, informative and easy to use to help in making decisions and not to make the user confused. Dashboards must show information effectively, focus on data and not on visual effects and use visual language that match human perception.

According to Few (2006), dashboards for organizations usually have many user groups with different information needs. Managers need views for comparison and diagnosing whereas teams need detailed information about everyday tasks. Well created dashboards have visual hierarchy, combine similar information together, and offer interactive elements like filtering and drill-down to grow participation. The use of colour must be intentional without cognitively overwhelming the user (Few, 2006, pp. 12, 26, 74).

When Power BI dashboards are integrated into SharePoint, they become part of the working environment (Microsoft 365 Documentation, n.d.). This integration grows the probability of seen analytics being applied in practice and contributes to the overall need of management based on information (Few, 2006, p. 12). Performance aspects such as effective data model and responsive visuals also affect the adoption of the system by the users and make them more confident (Few, 2006, p. 101).

4 Methods

This chapter goes through the methodological base and how the research was done. The task is to explain the reasoning for the chosen research strategy and show that the methodologies are fit for answering the research questions. It focuses on the development and implementation of an information system aimed at improving organizational goal management and analytical capacities. Because the goal is both practical and analytical, the methodology focuses on features of information systems research and applied development research. The methodological choices are consistent with development of a functional technical solution based on theoretical understanding.

4.1 Research approach

The research had a design-oriented approach, with the main outcome being a technical solution. This thesis creates a structured goal management system based on Microsoft cloud technologies. The technique is appropriate when the goal is not simply to study the theory but also to create a solution that meets a real-world organizational requirements. The research questions focus on how to develop a system that can replace spreadsheet-based solution while also supporting analytical decision making and organizational transparency.

The approach includes theory and practice also. The theoretical chapters show understanding of strategic goal management, data quality, and organizational information systems which will be used in the practical chapters. The practical chapters show how these ideas might then be used in system design and implementation. The solution is technically functional, also theoretically balanced and based on existing research.

4.2 Development methods

The system was created using an iterative approach, which is a process with many cycles of implementation, evaluation, and development. This way of working was chosen because organizational information systems must fit technical limitations and user requirements, which are many times understood only after prototypes have been tested in reality.

Many versions were developed and tested during the project and each version was shown to the commissioning organization representatives. Their feedback was collected each time and then used to edit it accordingly. This way usability issues and structural limits were found and fixed before the final solution was reached. The iterative process also helped the organization representatives to understand the system in depth as they saw the system grow and make recommendations for improvements. As a result, the finished system uses technological best practices within the found limitations and fits the needs of its users.

4.3 Data collection

The development was dependent on information given by the commissioning organization. This information given had details about sectors, teams, and organizational goals, as well as documents explaining how they manage organizational goals. The data was delivered in

excel format that could be directly imported into the platform used for implementation. It served as guidance for building the goal management system.

During the development process, the commissioning organization always gave qualitative feedback. Testing, review meetings, and demonstrations gave us ideas and information to use for the development process. This information was used to improve usability, as well as modify data structures and the interface design. The gathered information did not contain any personal or sensitive details and was solely for development purposes.

4.4 Process of designing the system

Requirements analysis is the first step in the system design process, which then moves on to development and validation. The first stage focused on understanding the organizations needs, current processes, and the limitations of the old Excel spreadsheet solution. This verified that the system is and would be used to handle real operational problems and not hypothetical needs.

The data architecture and system logic were studied and set at the second stage. Structured data models were created by setting the relationship between teams, sectors, and corporate goals. The final step had technical implementation using Microsoft cloud tools. The parts were designed and integrated to provide one platform that had data storage, automation, user interaction, and analytics. The final phase included testing and evaluation (Microsoft 365 Documentation, n.d.). The system was tested in realistic scenarios to make sure of dependability, correctness of data flows, and usability from two user roles, common users and management.

4.5 Evaluation methods

The system was reviewed using qualitative methods, including testing and validation together with the commissioning organization. Testing showed that data creation, filtering, and analytics worked as expected. The commissioning organization confirmed that the system met requirements and fit actual work processes. Evaluation was constant and each version of the system was reviewed and evaluated before developing the next version.

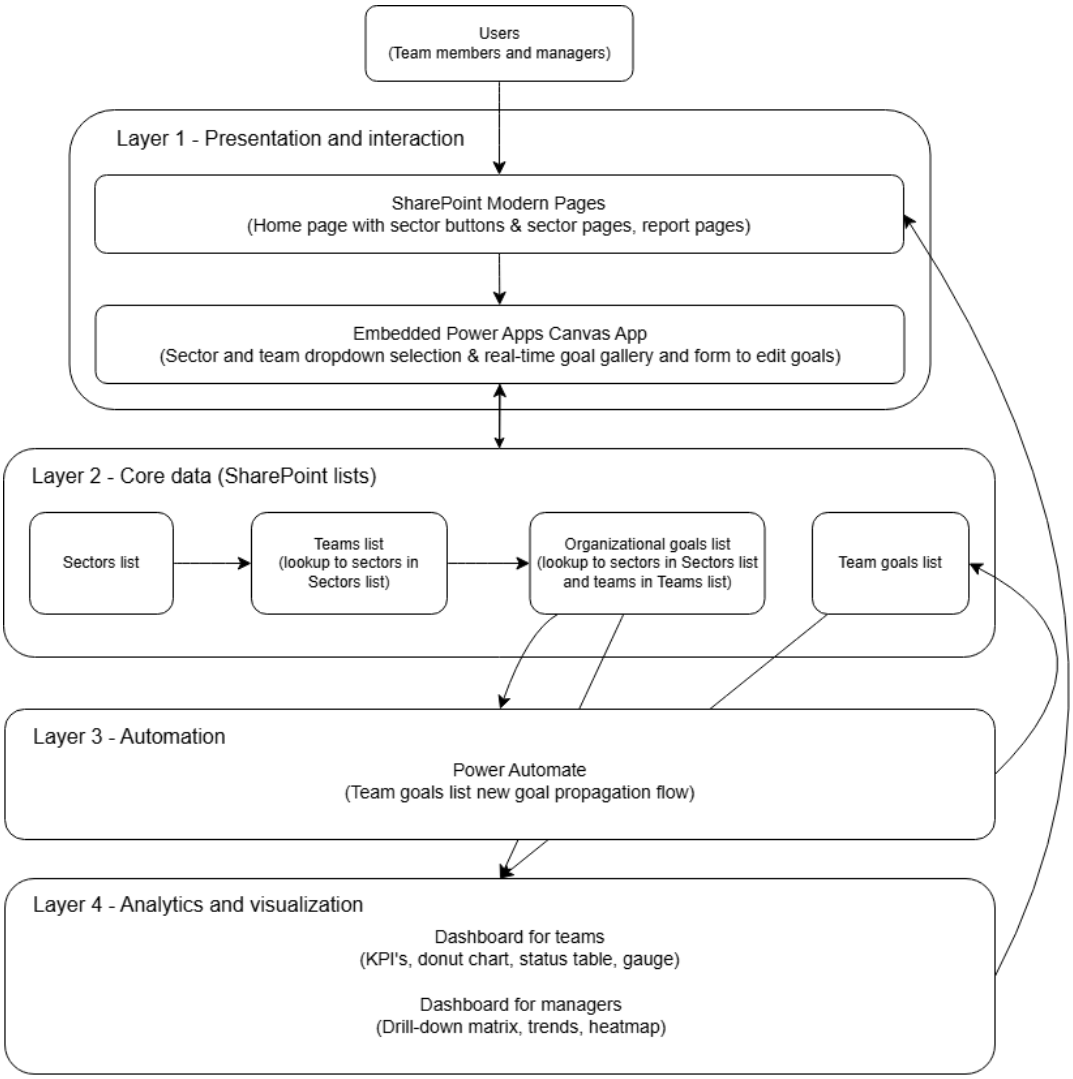
The evaluation focuses on three factors related to the research questions. The first objective was coherent and fully functioning system, defined as the system's ability to

replace spreadsheet-based goal management. The second objective was analytical usability, which shows if the system makes accurate data analysis and reporting possible. The third objective is organizational usefulness, which tells whether the solution grew transparency, usability and decision support or not. Based on testing sessions and user feedback showed that the final solution met every criteria and gave improvements over the previous Excel version.

5 Practical implementation of the goal management system

This chapter describes the technical process of creating the goal management system and how the conceptual framework mentioned previously was turned into an organizational goal management system. The solution was created for Kiipulasäätiö sr. to replace Excel spreadsheet goal monitoring with a digital platform which included analytics as well. The sections in the practical chapter go over the architecture (Figure 1), data modeling, automation logic, interface development, analytics, and security. Each part shows how the finished system meets each of the objectives.

Figure 1. Kiipula goal management system diagram



5.1 System architecture and platform selection

The main component of the solution was Microsoft SharePoint Online, which served as both the operational data layer and the user interface. We chose SharePoint because it can store data, manage permissions, and connect with other services in the same environment. SharePoint managed structured data and page content. Power Automate helped in keeping data between lists coherent and when appropriate, identical. Power Apps was used for dynamic interaction components that built on native SharePoint. Power BI was used for reporting and visualization.

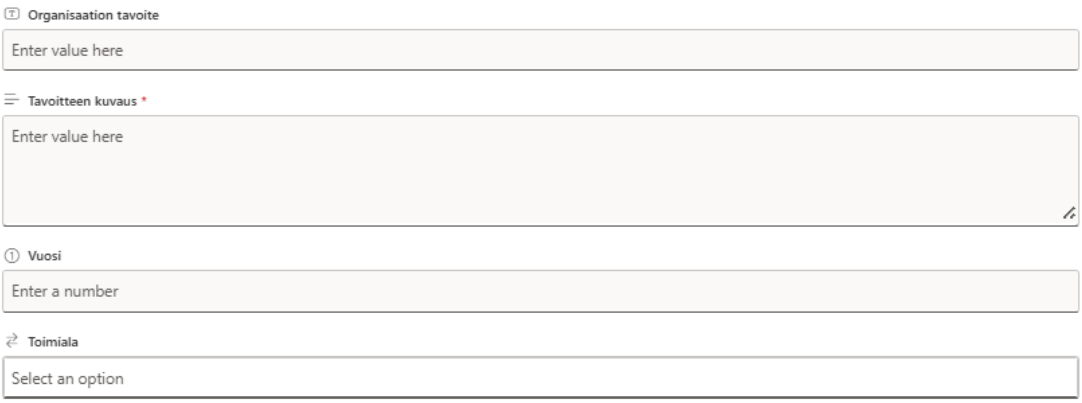
5.2 Data structure and SharePoint lists configuration

The data model served as the structural backbone of the system. It was built using interconnected SharePoint lists. The configuration process aimed at ensuring consistency in column types and validation rules. It aimed at ensuring the reliability of the data from the outset of the entry process. Four lists were designed. They represented the organizational sectors, teams, strategic goals, and team goals.

The entries in the list of sectors controlled the naming. It eliminated the discrepancies that existed in the spreadsheets. Information about the operational teams was included in the team list. It contained references to the respective sectors and supervisors. The lookup connections ensured that the teams were associated with existing sectors.

The high-level strategic goals were included in the list of organizational goals (Figure 2). Structured data contained the target year and the respective sector. A multi-line text field contained in-depth descriptions. The team goals list contained monitoring notes, completion dates, descriptive data for implementation planning, and the respective organizational goals. Default values and validation settings were implemented. They ensured the uniformity of the entries and reduced the possibility of errors.

Figure 2. View of adding an entry to the organizational goals list in SharePoint

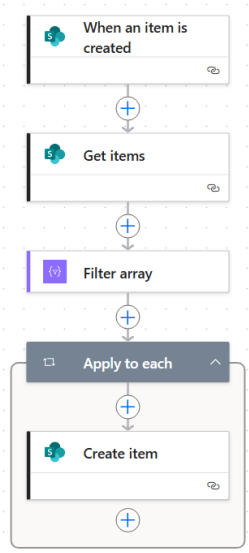


Hierarchical visibility within the organization was established via the connections between these listings. The system allowed for a direct examination of vertical alignment since each team goal was connected to its parent strategic target. In addition to ensuring that future growth could be supported without structural modification, this structure promoted analytical preparedness.

5.3 Automated goal propagation logic

When strategic goals were developed or altered, automation was implemented to ensure consistency and to do away with manual administrative labor. Every time a new organizational goal was saved, a procedure was set up (Figure 3) that caused the system to automatically recognize all teams in the chosen sectors and create goal entries at the team level. This procedure ensured that strategic goals were dispersed throughout the company as soon as they were developed.

Figure 3. Power Automate flow for automated goal entry propagation



Complete goal information, including sector goals, year of the goal, and description text, was retrieved by the automation logic. After that, it documented linked records for each team by matching these attributes with the team goal list. Important fields were filled in automatically, but others that needed team input were left blank to be filled up later.

The automated system decreased the possibility of errors that often come about with manual, human written and copying work. The result was a system where operational goals and strategic goals were consistent.

5.4 User interface design and navigation structure

Modern SharePoint pages were used to create the user interface, making a visually organized and user-friendly design possible. The goal was to lead users through goal

management tasks in a familiar way that followed their current workflow. The home page displayed organizational sectors (Figure 4). Users could select a target sector from a row of buttons. Once a sector was chosen, a page listed the teams under that sector. After selecting a team, the goals assigned to that team were shown.

Figure 4. Sector page buttons on the home page of the goal management system

The focus with design was put on consistency, visual hierarchy, and readability for ease of use. All pages had same styling, spacing, and alignment for a clean look. Navigation was made with search options and filtering dropdowns to make it easy to find wanted information. Completed goals were shown separately from active items, which made it possible for the users to see their progress fast.

Automatic customizing based on user identity was attempted in early versions. However it came about that the user directory data provided by the commissioning company was inaccurate and could not be reliably filtered. Consequently, the user interface was modified to allow users to manually select their sector and team. Despite adding a step to navigation, these improved clarity and prevented errors caused by erroneous profile data. Stakeholder feedback suggested that this is the method to use due to inaccurate user data.

They deemed that goals and analytics can be seen by the whole organization without restrictions.

5.5 Visual representation of goals

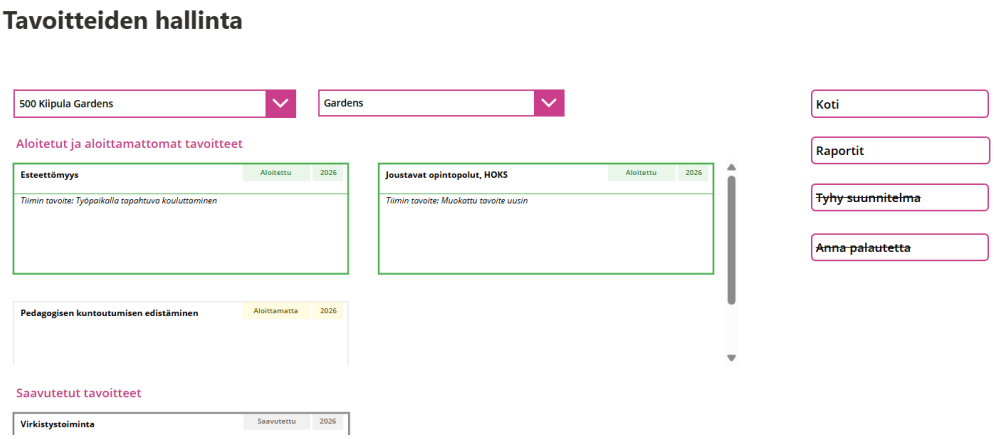
Embedded application components were developed to enable interactivity that native SharePoint pages could not provide. Early iterations relied on native SharePoint list views together with Quick Links buttons for sector and team selection. Team based or sector based filtering proved impossible with this approach. Native list views and card formatting did not support dynamic content refresh when a dropdown selection changed. Real time client side filtering of goals by selected team remained unachievable within SharePoint built in web parts.

To address these limitations the solution was redesigned in later development cycles. A custom Power Apps canvas application was embedded directly into SharePoint pages. The core interactive logic, including sector and team dropdowns, real-time goal filtering, and a gallery based display, was moved into Power Apps. When a user selects a sector and team from the dropdowns (Figure 5) the embedded app instantly filters and refreshes the displayed goals without any page reload. The result is a responsive application like experience inside the familiar SharePoint environment.

Figure 5. The sector and team selection dropdowns

In the final implementation each major sector received its own dedicated SharePoint page. These pages include the embedded Power Apps component with the correct sector value pre selected through a URL parameter. This means that users can skip to their relevant sector page by pressing buttons on the home page, after which they can simply select their specific team via dropdown (Figure 6). The use of sector specific pages, as well as filtering by specific teams, was a huge improvement in usability compared to the initial prototypes, as it reduces navigation steps.

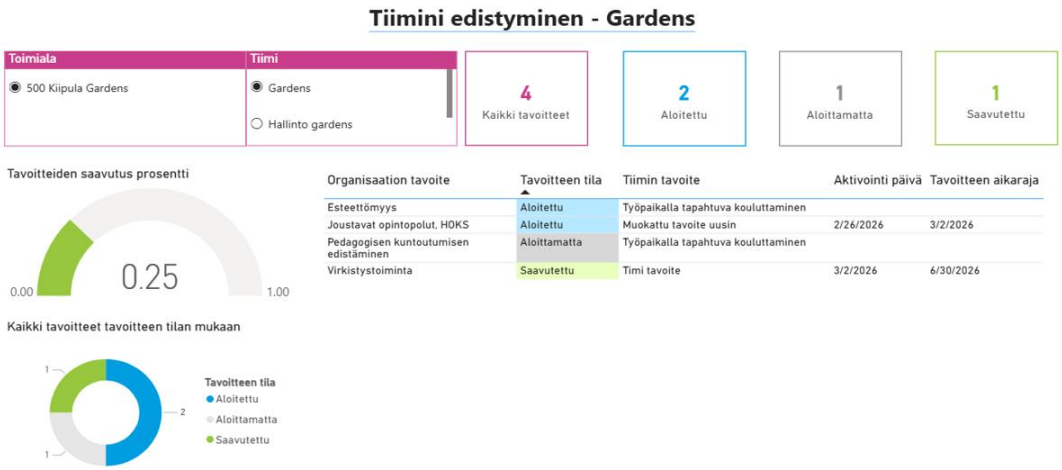
Figure 6. Team specific goal display provided when team was selected



5.6 Visualization design for general users

Visualizations intended for team members were designed to deliver straightforward and immediately understandable information about the progress of their own team goals. A dedicated report page (Figure 7) has been designed within the Power BI environment. At the top of the report page, the user found key performance indicators that showed the total number of goals, the number of goals completed, goals in progress, and goals that have yet to be started. This provided the user with an instant overview of the situation.

Figure 7. General workers Power BI report



A donut chart showed the distribution of goals in the three categories of 'not started,' 'in progress,' and 'completed'. A table holds all team goals had columns for organizational

goal, the current status of the goal, the team's target, the activation date, and the deadline. Conditional formatting was used for the status column, so that blue background was used for 'in progress' goals, grey for 'not started' goals, and green for 'completed' goals.

A clustered column chart visualizes the number of goals manually activated each month. This setup helps team members track whether goal activation is progressing steadily over time. All visuals are updated automatically based on the sector and team selected in the dropdown at the top of the report. The overall design is simple and clear, for users to understand their team's process without needing advanced analytical skills.

5.7 Visualization design for managerial oversight

Reports intended for managerial users provide a comprehensive analytical perspective covering the entire organization. A separate report page named Leadership Dashboard was developed in the same Power BI file. High-level summary cards display total goal counts, completion rates, goals in progress, goals not started and overdue goals across all sectors and teams.

The main visual was a drillable matrix that displayed the data by sector which was followed by each team within that sector. It displayed the percentage completed, total goals, completed goals, overdue goals, and average goals per team. This made it easy to see the difference in performance between teams. Managers could expand the sector section to view the teams within each sector or collapse it to view the overall information.

There was also a line chart showing the trends of completed and in-progress goals over time, a heat map matrix showing the percentage of completed goals, and a table showing teams that need assistance. Another visual was the decomposition tree showing the percentage of goals completed by sector, team, and organizational goal.

Lastly, the what-if parameter was used by managers to input a percentage that was desired to be completed. It then displays the difference between the current state and the desired state.

5.8 Integration of analytics into the intranet environment

For the analytics to be accessed within the same workspace as the daily activities, the reports were integrated into the intranet. The creation of different pages for the managers and the operational users was done to meet the different needs for the information. The embedding of the reports made it unnecessary for other applications to be used for viewing the analytics often.

Two different types of users could access the two different pages, each with the permission set for the users. The managers could access the analytics for the entire organization, and the operational users could access the team reports. By continuous testing access control, data refresh, and page loading it was shown that everything worked with different user accounts.

5.9 Access control and security design

Data protection and usability were matched in security design. Two primary user groups were identified. Administrative users and managers were granted full permissions, giving them rights to control analytics, automation, lists and site settings. The rights provided to general users were limited to viewing and interacting with goal submissions and normal pages. The implementation of strict access control for teams was not feasible without risking deleting legitimate users, as the information provided in the directory data was not accurate in representing all team members.

Considering corporate policy and interface best practices, it was decided that this solution would provide general users access to all SharePoint pages except the manager reports page. After reviewing this solution, it was determined that there were minimal risks involved, as there was no financial or personal information contained in the data. The version history of changes could be monitored, providing an added layer of protection.

5.10 Iterative development and validation

Development went through many cycles, demonstration, and modifications. First prototypes brought about organizational and technical limitations that made redesigning several aspects unavoidable. The most severe need for redesign surfaced when it was revealed that lacking user information from the commissioning organization would lead automating

personalization based on user profiles to fail. As a result, the system was redesigned to depend on team selection, still having transparency and dependability.

Several versions of the platform were created during the process. Every realistic scenario, like setting strategic goals and tracking the progress, were used to test each version. Logic, design, and navigation were all improved based on the feedback given by the commissioning organization. By using this strict process the final system met user expectations and performed reliably in needed situations.

6 Results

Using the goal management system at Kiipulasäätiö sr. has shown more clean data, a clearer process of managing the goals, and better analysis possibilities. The newly developed solution has replaced the traditional use of spreadsheets through a connected network linking operational tasks directly to strategic objective goals. This has addressed the inefficient way by which progress tracking and completion of analyses were previously conducted.

Through the application of a single data model, the variations in the terminology used to collect and store information, both in terms of use and in formatting the collected and stored information, have been minimized. All organizational objectives are now within a single consistent standard, predetermined data entry fields, with automated validation criteria, thereby allowing trustful analyses on how well the organization performs. The system reduced manual work by replacing repetitive tasks with automations. The goal creation logic made sure that strategic objectives were communicated quickly to the right teams, minimizing errors and delays.

Feedback and testing meetings with the commissioning organization showed that staff could correctly use the system with minimal training. The users were able to easily switch from selecting sectors to viewing information specific to their own team and updating goals. Info texts and color coding helped users in understanding the information shown and keeping their knowledge up to date. By continuously using the tool, the number of incomplete or outdated records has been reduced, improving the quality of the data collected.

Operational data flows from the operational data source to the visual dashboards built with Power BI to provide trend lines to evaluate and compare the progress against the performance. By doing so, management can review organizational trends and compare each department's performance. Team members can view the trends and quickly see any gaps between themselves and the rest of the organization. This made it easier for managers to find things that needed more attention or resources and to make decisions based on evidence.

The security was redesigned and then met while staying within the user data constraints. The methodology for combining permission groups and balancing the development of user permissions with risk management was not able to be fully demonstrated because the user directories of the commissioning organization were not updated. All changes could be reversed and traced thanks to version tracking. Since no sensitive financial or personal data was in the system, the commissioning organization determined that the risks were not significant. This confirmed that the chosen configuration was appropriate for the operational environment.

Overall, the results indicate that goal management can improve by using an integrated digital platform instead of spreadsheets. The solution effectively addressed the research problem by providing a scalable and organized framework that links operational tasks with strategic planning while offering analytical insights.

7 Summary

The goal of this thesis was to create a structured goal management system that would improve the connection between the organization's strategic plan and the day to day operations of the organization. It studied theories on strategic management, the alignment of goals, and the quality of data to build a solution fit for the purpose. Three main research questions regarding the replacement of current Excel spreadsheet method, the providing of assistance in making analytical decisions, and the development of transparency and usability of data were targeted during the study.

Ultimately, the theoretical analysis showed the need for systems that can connect higher level goals with practical activities in order to implement strategies successfully. Based on previous research findings, organizations often have difficulty in transforming their strategy into measurable actions for the teams because data is stored in multiple locations. These

findings provided functional requirements that the solution needed to meet as well as a theoretical framework to guide the designing of the system.

The hands-on training showed how to implement a cloud system to meet all these objectives. A structured data architecture was developed to provide a standard approach to data for both consistency and analytical readiness. Automation techniques were used to synchronize data at all levels of the organization. All parts of an interactive user interface were created to promote interaction and use. The purpose of analytical dashboards was to provide views into operational data that could be used for action. Prototypes were iteratively tested and improved which made sure that the final system achieved both the actual organizational needs and technical best practices.

The results of the evaluation showed that the system goals were achieved, and the end result was a unified platform that supported trust in data management and analysis, instead of manual processes. Managers were able to access detailed information, and users reported that there is now clearer idea of purpose and responsibility for them. These results show that integrated information systems are important in improving organizational learning and alignment with strategic goals.

The thesis makes scholarly and practical contributions offering a usable solution that the commissioning organization can use and expand. The thesis also provides an example of how an applied system design could be built on theoretical concepts of information systems and goal management. This study shows the fact that theory and the implementation of technologies work successfully together when implemented at the same time.

In the future, the system could be further developed through the addition of automated deadline alerts, the use of prediction analytics, or the addition of more organizational data sources to the existing database. These improvements would raise the analytical value of the data being used as well as improve the working motivation. However, even without these improvements, the current implementation provides a strong foundation and shows improvement compared to the previous method used.

The thesis concludes that an effectively designed information system can fix the gap between operations and strategy. The system created in this way provides transparent goal management through the integration of analytics with automated systems, user-centered design of software, and structured data architecture.

References

Baptista, J., Chestney, R., Goebel, N., McConell, J., Marsh, E., Miller, P., Pohl, A., Robertson, J., & White, M. (n.d.). *From intranet to digital workplace: How to evolve your strategy*. 38.

Few, S. (2006). *Information dashboard design: The effective visual communication of data*. Beijing ; Cambride [MA] : O'Reilly. http://archive.org/details/pdfy--fQ3cC8TeDUArgti

Kaplan, R. S., & Norton, D. P. (2001). *The strategy focused organization*.

Locke, E. A., & Latham, G. P. (2002). Building a practically useful theory of goal setting and task motivation: A 35-year odyssey. *American Psychologist*, *57*(9), 705–717. https://doi.org/10.1037/0003-066X.57.9.705

Microsoft 365 documentation. (n.d.). Microsoft 365 Documentation. Retrieved 11 February 2026, from https://learn.microsoft.com/en-us/microsoft-365/?view=o365-worldwide

Mintzberg, H. (1994). The Fall and Rise of Strategic Planning. *HARVARD BUSINESS REVIEW*, 105–114.

Panko, R. R. (2005). *What We Know About Spreadsheet Errors*. 33.

Popovič, A., Hackney, R., Coelho, P. S., & Jaklič, J. (2012). Towards business intelligence systems success: Effects of maturity and culture on analytical decision making. *Decision Support Systems*, *54*(1), 729–739. https://doi.org/10.1016/j.dss.2012.08.017

Simons, R. (1994). *Levers of Control*. Harvard Business Press.

Verma, P., & Sharma, R. R. K. (2016). *Structure, Evaluation System, and Employee Resistance: Impediments to Horizontal Strategy Implementation in Diversified Corporations*. 8.

Wang, R. Y., & Strong, D. M. (1996). Beyond Accuracy: What Data Quality Means to Data Consumers. *Journal of Management Information Systems*, *12*(4), 5–33.

Williams, S. P., & Schubert, P. (2018). Designs for the Digital Workplace. *Procedia Computer Science*, *138*, 478–485. https://doi.org/10.1016/j.procs.2018.10.066

Appendix 1: Data management plan

The methods for handling the research data produced during the thesis project, "Microsoft Based performance goal management system for strategy execution," are described in this Data Management Plan (DMP). This strategy complies with the recommendations made by HAMK University of Applied Sciences.

Description of thesis research data

The creation and deployment of a Microsoft-based goal management system is the main focus of this thesis. Thus, the majority of the research data is technical and qualitative, with an emphasis on system settings, design decisions, and user input.

- Data collection techniques include:
  - System design and configuration documentation includes Power BI dashboard designs, Power Automate flows, Power Apps components, SharePoint list structures, and architectural decisions.
  - Monitoring how the system behaves as it is being developed and tested.
  - Getting opinions from the commissioning organization while developing and validating iteratively.
  - Examining Microsoft documentation and existing literature about goal management, SharePoint, and Power Platform.
- Target group:
  - Kiipulasäätiö sr., the commissioning organization, and its staff members who will be utilizing the created system are involved in the study. No personally identifiable information will be gathered or handled.
- Forms of the data:
  - Markdown or plain text documents that provide configurations, design justifications, and system specifications.
  - Exported configurations (such as JSON, XML, or comparable configuration files) from Power Automate, Power Apps, SharePoint, and Power BI.
  - Screenshots and images of the features and user interface of the system.

Management and storage of the research data

The thesis author's password-protected computer will be used to store and process the research data. A specific project folder will house all project-related items, such as configuration files, system manuals, and any feedback gathered.

In order to guarantee data security and privacy, the following actions will be taken:

- Only the thesis author will have access to the project folder and its contents. Access to pertinent documentation for review purposes may be provided to the thesis supervisor.
- The project folder will be regularly backed up and stored on an external, encrypted storage device that is physically isolated from the main computer.
- Unless specifically authorized by HAMK and the commissioning organization, and only if the service satisfies the necessary security requirements, no sensitive or private data will be kept in cloud services. It is not anticipated that direct, private organizational data will be included in the research data due to the nature of the project (system development).
- Since the data mainly relates to the technical execution of the system and is not sensitive operational data, no particular agreements have been reached with the commissioner about data treatment beyond the general thesis rules.

Processing of personal data and sensitive data

This thesis research will not gather or process any sensitive personal information. The goal management system's functionality and technological implementation are the main focus, not the performance or private data of specific users. Any input received will be generalized and anonymised with an emphasis on the efficacy and usefulness of the system.

Because no personal information is being gathered, research participants do not require a privacy notice.

Ownership of research data

According to HAMK's general thesis guidelines, the thesis author and HAMK University of Applied Sciences will be the owners of the research data produced during this thesis project, including the system design documentation, configuration files, and any technical assets developed.

- Copyrights: Since the system is being created for the commissioning organization, Kiipulasäätiö sr., the intellectual property rights for the system and its components will be managed in accordance with their agreement. Kiipulasäätiö sr., HAMK, and the thesis author will enter into a separate agreement pertaining to the ownership and use of the developed system.

- Information gathered by others: All outside information or documents (such as Microsoft documentation) will be appropriately referenced in accordance with HAMK's citation standards, and their copyrights will be upheld.

Further use of research data after the completion of the thesis

After the thesis is finished, the study data that mostly consists of system documentation and configuration details will not be used again or made publicly available.

For a year after the thesis is approved, the data will be safely stored by the thesis author so that the findings can be checked if needed. The data will be safely erased after this time. The commissioning organization will maintain and operate the established system as it was implemented for Kiipulasäätiö sr. in accordance with their internal regulations and a separate thesis agreement.